

Plate Reader — Command Reference

The plate reader measures a 96-well plate in two modes:

- **Luminescence:** returns a light-intensity reading for each well.
- **Absorbance:** uses a selected wavelength and returns two values per well, which you divide to obtain the result.

The reader must be closed before taking a measurement.

Connecting and sending commands

Connect to the USB serial port at **115200 baud, 8N1, no flow control**. Clear any startup text before sending the first command.

Send one command, then wait for its complete response before sending another. Allow up to one second for a response.

The examples below show bytes in hexadecimal. Every command has this form:

```
A7 03 [command] 00 [parameter: 2 bytes] 00 FF
```

Use `00 00` when a command has no parameter. For a two-byte number, send the high byte first: for example, 450 is `01 C2`.

Responses begin with `A7 83`, followed by the command code and a status byte. Status `00` means the command succeeded. Responses end in `00 FF`; these are fixed bytes, not a checksum.

Opening and closing

Use **OPEN (07)** to open the reader and **CLOSE (06)** to close it. Measurements are unavailable while it is open.

```
Open:      A7 03 07 00 00 00 00 FF
Receive:   A7 83 07 00 00 00 00 FF

Close:     A7 03 06 00 00 00 00 FF
Receive:   A7 83 06 00 00 00 00 FF
```

Opening and closing preserve the measurement settings.

Measurement settings

Wavelength — SET_WAVELENGTH (01)

For absorbance, select a wavelength from **340 to 850 nm**. This example selects **450 nm** (01 C2):

```
Send:      A7 03 01 00 01 C2 00 FF
Receive:   A7 83 01 00 01 C2 00 FF
```

The response repeats the accepted wavelength. Wavelength does not affect luminescence.

Gain — SET_GAIN (31)

Gain is a signal multiplier from **1 to 16** and defaults to **1** at startup and after RESET. Send SET_GAIN before either measurement, even when using the default. This example selects **1**:

```
Send:      A7 03 31 00 00 01 00 FF
Receive:   A7 83 31 00 00 01 00 FF
```

Higher gain increases the reported readings. The response repeats the accepted gain.

For absorbance, set wavelength and gain before reading. For luminescence, set gain; wavelength is not used.

Measurements

Both modes return all 96 wells in this order:

```
A1, A2, ..., A12, B1, B2, ..., B12, ..., H12
```

Each returned number occupies two bytes, high byte first. For example, 02 B2 is a reading of 690. Repeated measurements with the same settings return identical values.

Luminescence — READ_LUMINESCENCE (20)

```
Send:      A7 03 20 00 00 00 00 FF
Receive:   A7 83 20 00 [96 readings] 00 FF
```

The response is **198 bytes**. The readings are in relative light units (RLU) and can be used directly.

Absorbance — READ_ABSORBANCE (21)

```
Send:      A7 03 21 00 00 00 00 FF
Receive:   A7 83 21 00 [96 A values] [96 B values] 00 FF
```

The response is **390 bytes**. It contains all A values first, then all B values. For each well, divide its A value by its B value:

```
Result = A / B

Example: 855 / 1765 = 0.4844
```

Use decimal division. Each well has its own B value. Gain scales A; B is unchanged. The bracketed parts in these examples represent data, not literal text.

Reader state — GET_STATE (02)

Use this command to see the measurement settings and whether the reader is open or closed.

```
Send:      A7 03 02 00 00 00 00 FF
```

The **12-byte** response is arranged as follows:

```
A7 83 02 00 [wavelength: 2 bytes] [gain: 2 bytes] [wavelength set] [open] 00 FF
```

Field	Meaning
Wavelength	Wavelength in nm; 00 00 = not set
Gain	Current gain; defaults to 00 01
Wavelength set	00 = not set; 01 = set
Open	00 = closed; 01 = open

For example, with the default settings:

```
Open:      A7 03 07 00 00 00 00 FF
Receive:    A7 83 07 00 00 00 00 FF
Get state:  A7 03 02 00 00 00 00 FF
Receive:    A7 83 02 00 00 00 00 01 00 01 00 FF

Close:     A7 03 06 00 00 00 00 FF
Receive:    A7 83 06 00 00 00 00 FF
Get state:  A7 03 02 00 00 00 00 FF
Receive:    A7 83 02 00 00 00 00 01 00 00 00 FF
```

The open field changes from 01 to 00. The other fields stay the same.

Reset — RESET (04)

RESET clears wavelength, and restores gain 1. It leaves the reader's open/closed position unchanged.

```
Send:      A7 03 04 00 00 00 00 FF
Receive:    A7 83 04 00 00 00 00 FF
```

After resetting, set gain before either measurement; for absorbance, set wavelength too. Disconnecting and reconnecting USB does not reset the reader.

Error responses

An error response is always **8 bytes**. The fourth byte tells you what went wrong:

```
A7 83 [command] [error] 00 00 00 FF
```

Error	What to do
01 — Unknown command	Check the command code.
02 — Invalid parameter	Check the allowed range, or use 00 00 for a command without a parameter.
05 — Missing settings	Send SET_WAVELENGTH and SET_GAIN.
06 — Invalid request	Check that the fixed bytes match the examples.
07 — Reader open	Send CLOSE before reading.

A failed command does not change the reader's settings.

A response may arrive in several pieces. Keep receiving until you have its full length; do not stop at the first FF, because that byte can also occur in data. Check the status first: a failed read returns an 8-byte error, not plate data. If a request was incomplete or no response arrives, wait at least 250 ms before retrying.